AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Electrical and Electronic Engineering

Course No: EEE 4154

Course Name: Power Systems II Lab

A Software Project on Power Flow Using PowerWorld Simulator

Submitted by

Eyenun nishat

ID: 20220105237

Year: 4th Semester: 1st

Section: D2

Objective: The objective of this assignment is to analyze power flow in a given power system using the PowerWorld Simulator and obtain the load flow and fault info.

calculation id : 20220105237

System Base Quantities:

Sbase = 100 +37 = 137 MVA

Vbase = 13.8+7=20.8 kV in the zone of G1

For tx line ,

V[lbase]=(230*20.8)/13.8=346v

Z base= (346)^2/137= 873 ohm

For line 1:

Z pu= (0.08 + j0.5) x15/873=1.37+ 8.5*10^-4

Y pu= {3.4x10^(-6)j x15}/ 1.14*10^-3 =0.041

For line 2: z pu = (0.08 + j0.5) x25/873= 2.29*10^-3+ 0.01j .

Y pu= {3.4x10^(-6)j x25}/ 1.14*10^-3 = 0.07 pu

For line 3: for 40 km, z pu= 3.6*10^-3 + 0.0236j and Y pu= 0.12pu

Line 4: line = 15 km

Z pu= (0.08 + j0.5) x15/873=1.37+ 8.5*10^-4

Y pu= {3.4x10^(-6)j x15}/ 1.14*10^-3 =0.041

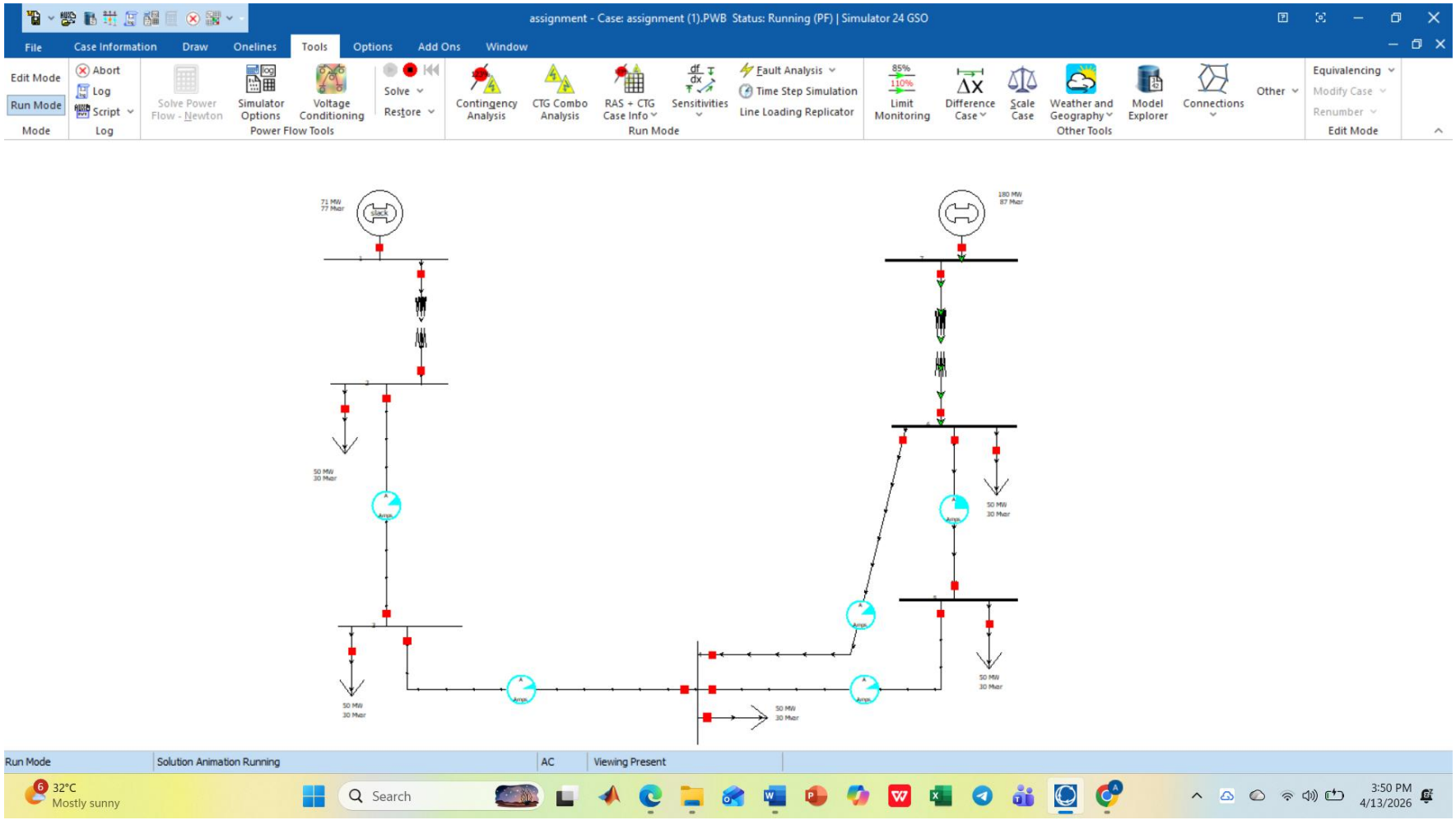
Line 5: line = 50 km

Z pu= (0.08 + j0.5) x50/873=4.5*10^-3+0.02i

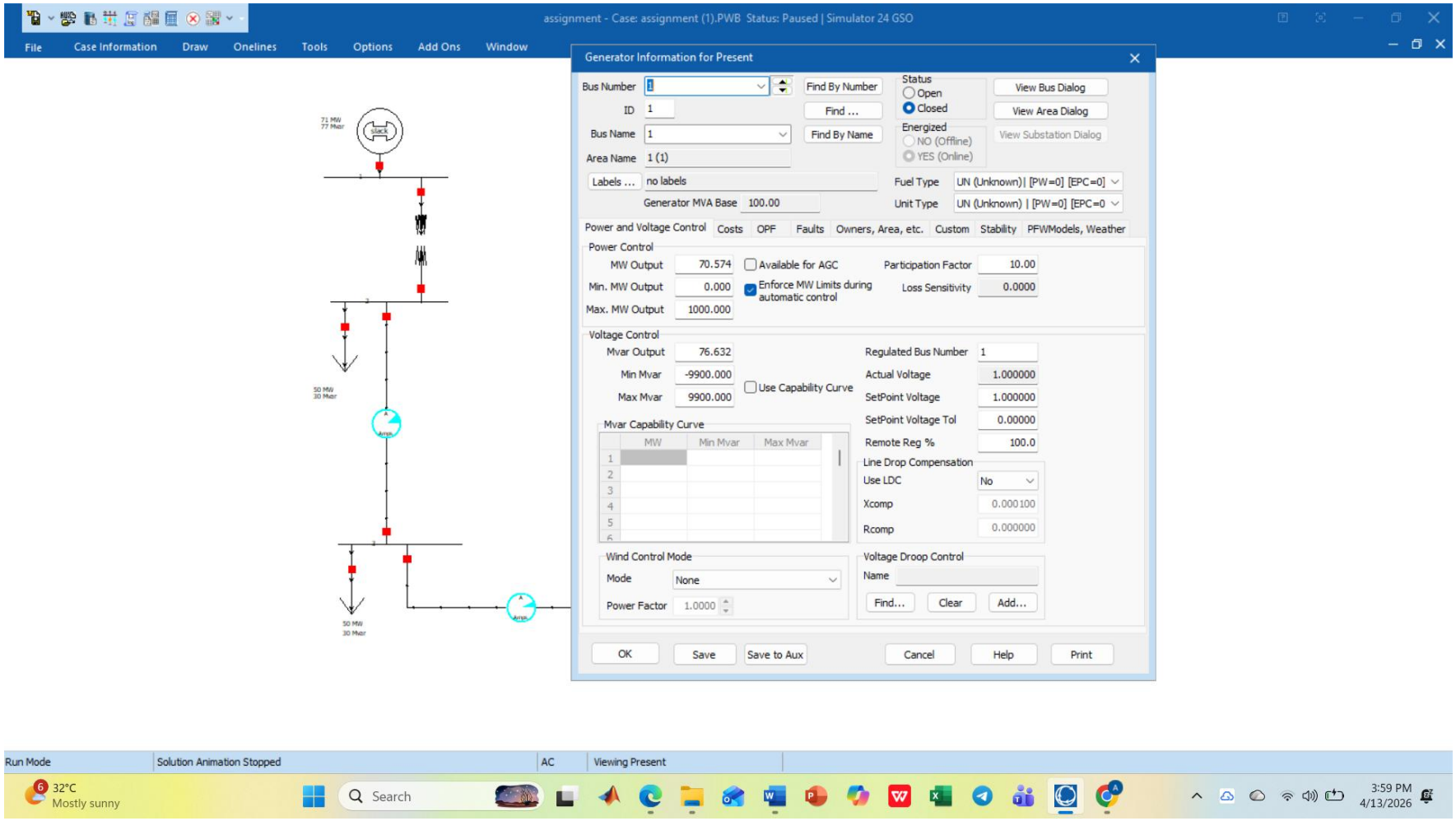
Y pu= {3.4x10^(-6)j x50}/ 1.14*10^-3=0.16 pu

Sample Process Drawing

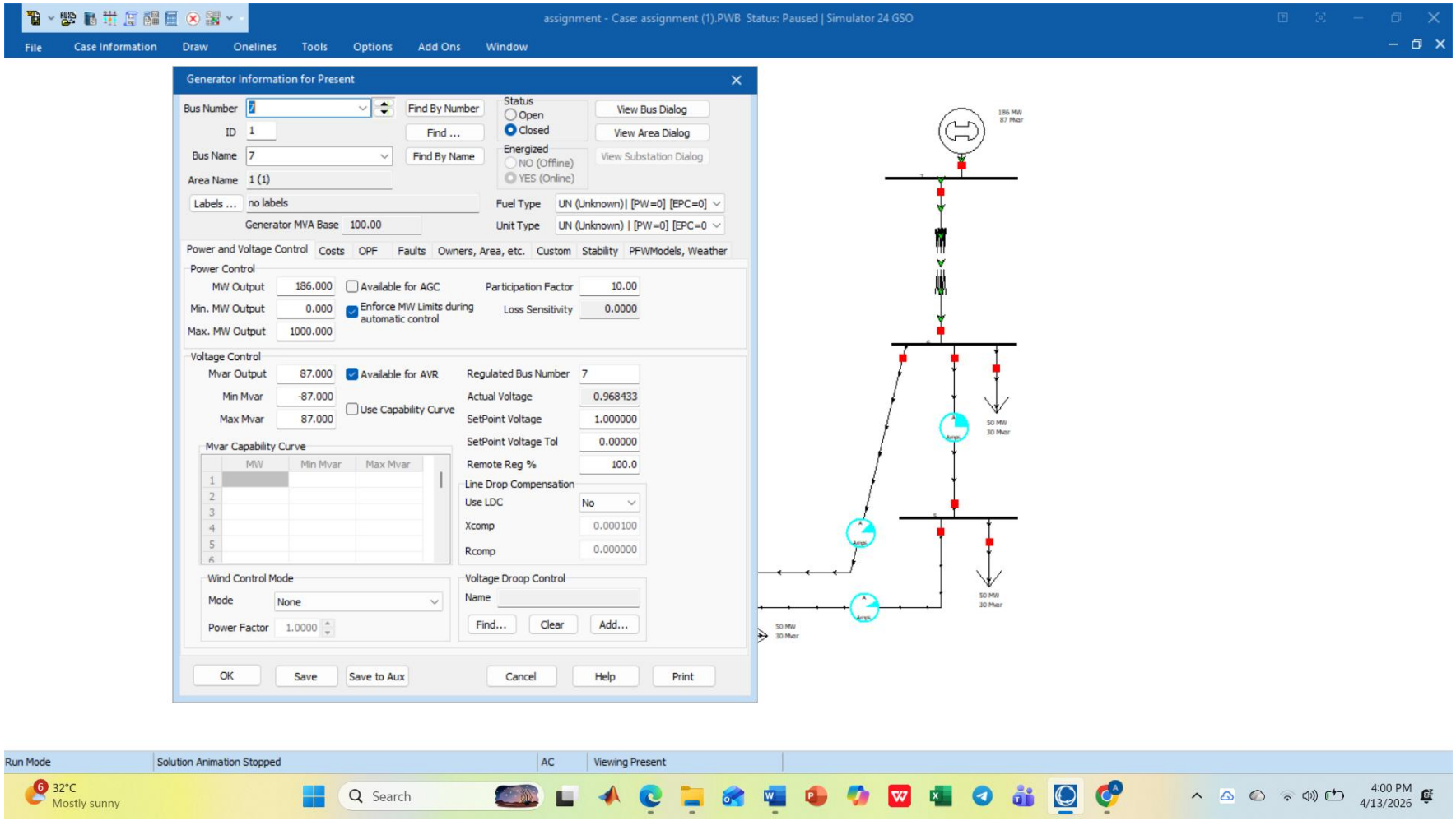
Swing Bus Settings:



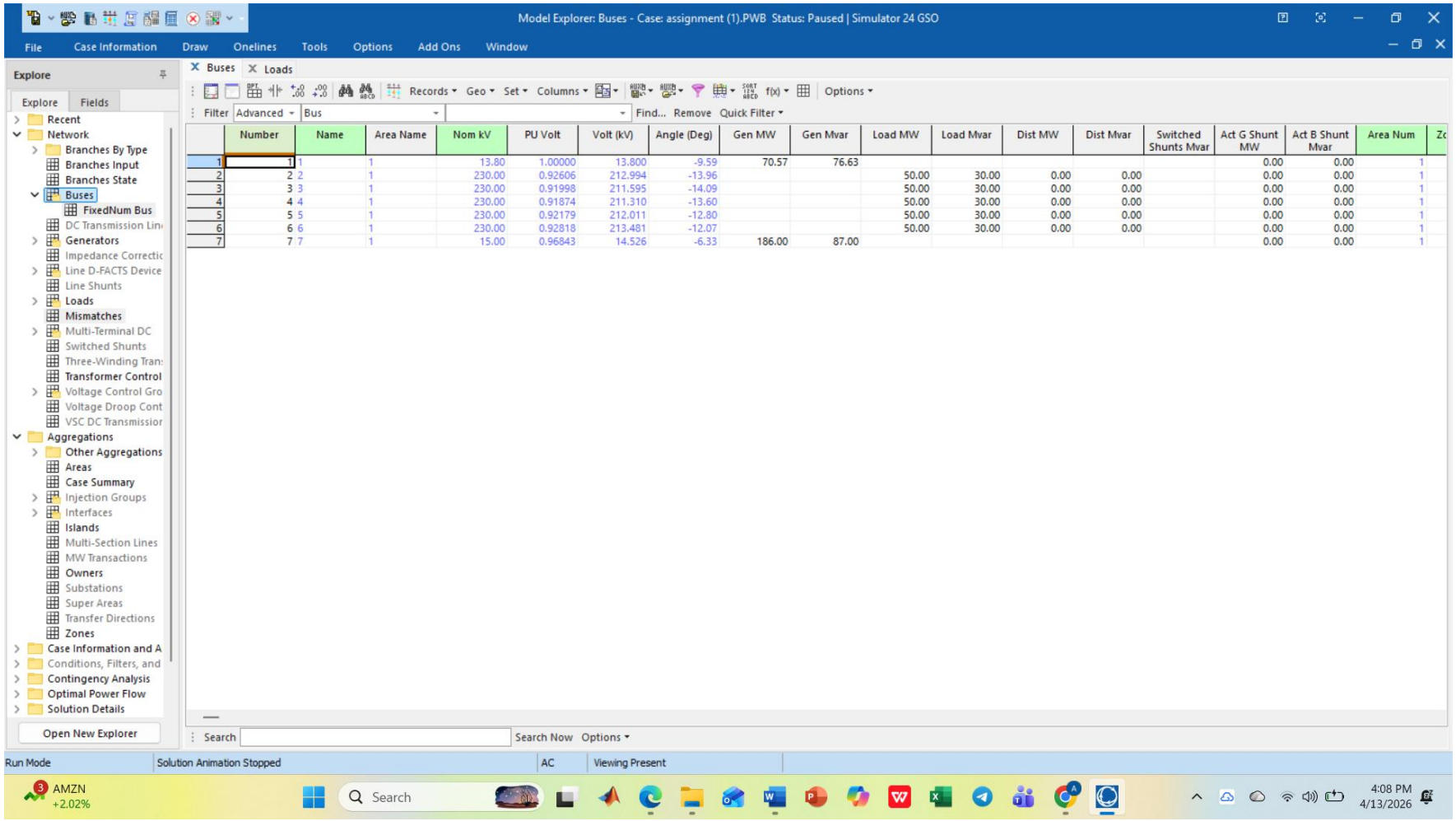
Generator-1 Settings:



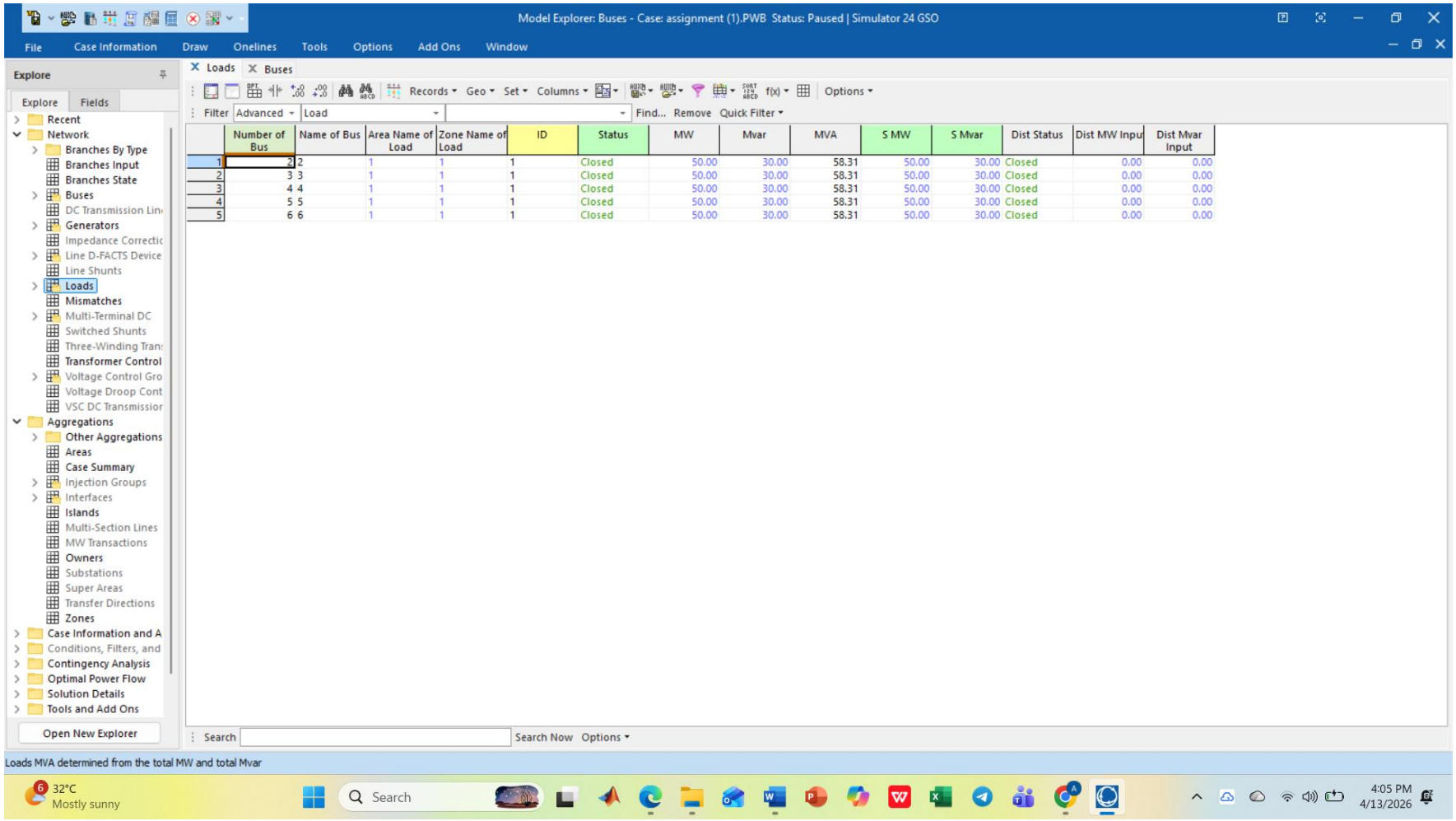
Generator-2 Settings:



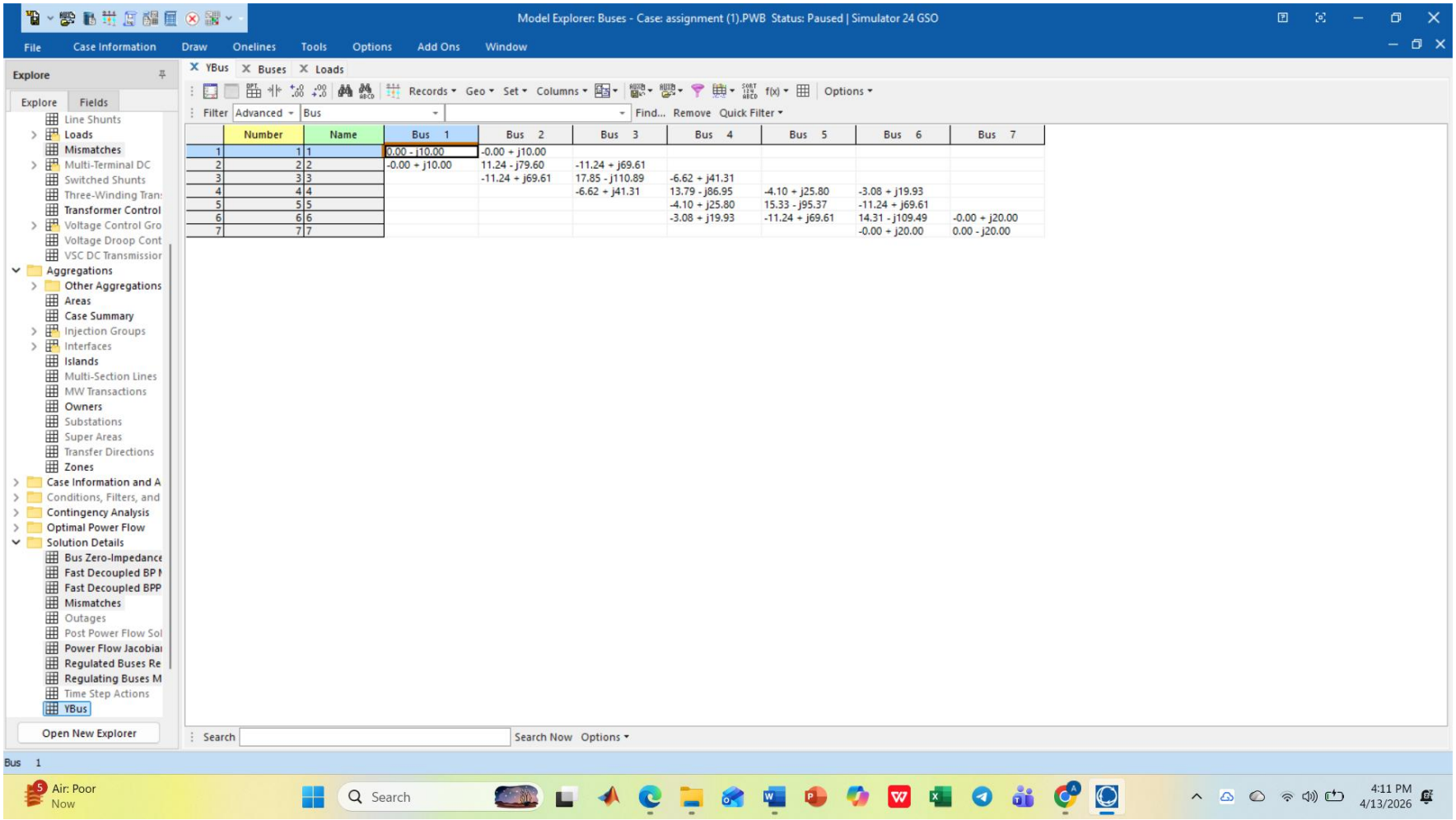
Bus input :


Model Explorer: Buses - Case: assignment (1).PWB Status: Paused | Simulator 24 GSO
File
Case Information
Draw
Onelines
Tools
Options
Add Ons
Window
Explore
Explore
Fields
Recent
Network
Branches By Type
Branches Input
Branches State
Buses
FixedNum Bus
DC Transmission Lin
Generators
Impedance Correctic
Line D-FACTS Device
Line Shunts
Loads
Mismatches
Multi-Terminal DC
Switched Shunts
Three-Winding Tran
Transformer Control
Voltage Control Gro
Voltage Droop Cont
VSC DC Transmissior
Aggregations
Other Aggregations
Areas
Case Summary
Injection Groups
Interfaces
Islands
Multi-Section Lines
MW Transactions
Owners
Substations
Super Areas
Transfer Directions
Zones
Case Information and A
Conditions, Filters, and
Contingency Analysis
Optimal Power Flow
Solution Details
Open New Explorer
Buses
Loads
Records
Geo
Set
Columns
Options
Filter
Advanced
Bus
Find...
Remove
Quick Filter
Number
Name
Area Name
Nom kV
PU Volt
Volt (kV)
Angle (Deg)
Gen MW
Gen Mvar
Load MW
Load Mvar
Dist MW
Dist Mvar
Switched Shunts Mvar
Act G Shunt MW
Act B Shunt Mvar
Area Num
1 1 1 1 13.80 1.00000 13.800 -9.59 70.57 76.63 0.00 0.00 1
2 2 2 1 230.00 0.92606 212.994 -13.96 50.00 30.00 0.00 0.00 0.00 0.00 1
3 3 3 1 230.00 0.91998 211.595 -14.09 50.00 30.00 0.00 0.00 0.00 0.00 1
4 4 4 1 230.00 0.91874 211.310 -13.60 50.00 30.00 0.00 0.00 0.00 0.00 1
5 5 5 1 230.00 0.92179 212.011 -12.80 50.00 30.00 0.00 0.00 0.00 0.00 1
6 6 6 1 230.00 0.92818 213.481 -12.07 50.00 30.00 0.00 0.00 0.00 0.00 1
7 7 7 1 15.00 0.96843 14.526 -6.33 186.00 87.00 0.00 0.00 1
Search
Search Now
Options
Run Mode
Solution Animation Stopped
AC
Viewing Present
AMZN
+2.02%
Search
4:08 PM
4/13/2026

Load Settings:

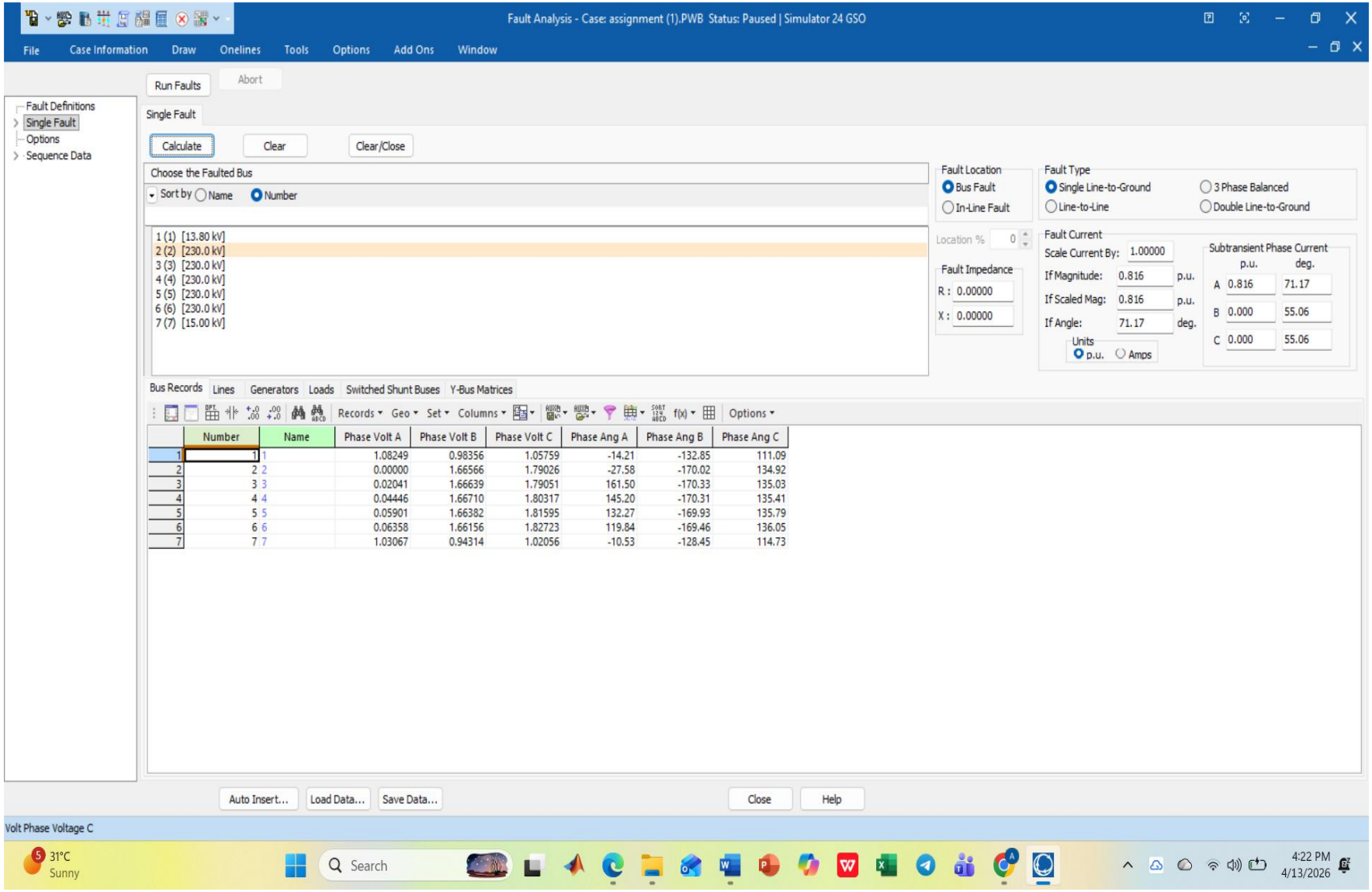


Ybus Matrix :


Model Explorer: Buses - Case: assignment (1).PWB Status: Paused | Simulator 24 GSO
File
Case Information
Draw
Onelines
Tools
Options
Add Ons
Window
Explore
Explore
Fields
Line Shunts
Loads
Mismatches
Multi-Terminal DC
Switched Shunts
Three-Winding Tran
Transformer Control
Voltage Control Gro
Voltage Droop Cont
VSC DC Transmissior
Aggregations
Other Aggregations
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Case Summary
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Multi-Section Lines
MW Transactions
Owners
Substations
Super Areas
Transfer Directions
Zones
Case Information and A
Conditions, Filters, and
Contingency Analysis
Optimal Power Flow
Solution Details
Bus Zero-Impedance
Fast Decoupled BP M
Fast Decoupled BPP
Mismatches
Outages
Post Power Flow Sol
Power Flow Jacobiar
Regulated Buses Re
Regulating Buses M
Time Step Actions
YBus
Open New Explorer
YBus
Buses
Loads
Records
Geo
Set
Columns
Options
Filter
Advanced
Bus
Find...
Remove
Quick Filter
Number
Name
Bus 1
Bus 2
Bus 3
Bus 4
Bus 5
Bus 6
Bus 7
1 1 1 0.00 - j10.00 -0.00 + j10.00
2 2 2 -0.00 + j10.00 11.24 - j79.60 -11.24 + j69.61
3 3 3 -11.24 + j69.61 17.85 - j110.89 -6.62 + j41.31
4 4 4 -6.62 + j41.31 13.79 - j86.95 -4.10 + j25.80 -3.08 + j19.93
5 5 5 -4.10 + j25.80 15.33 - j95.37 -11.24 + j69.61
6 6 6 -3.08 + j19.93 -11.24 + j69.61 14.31 - j109.49 -0.00 + j20.00
7 7 7 -0.00 + j20.00 0.00 - j20.00
Search
Search Now
Options
Bus 1
Air: Poor
Now
Search
4:11 PM
4/13/2026

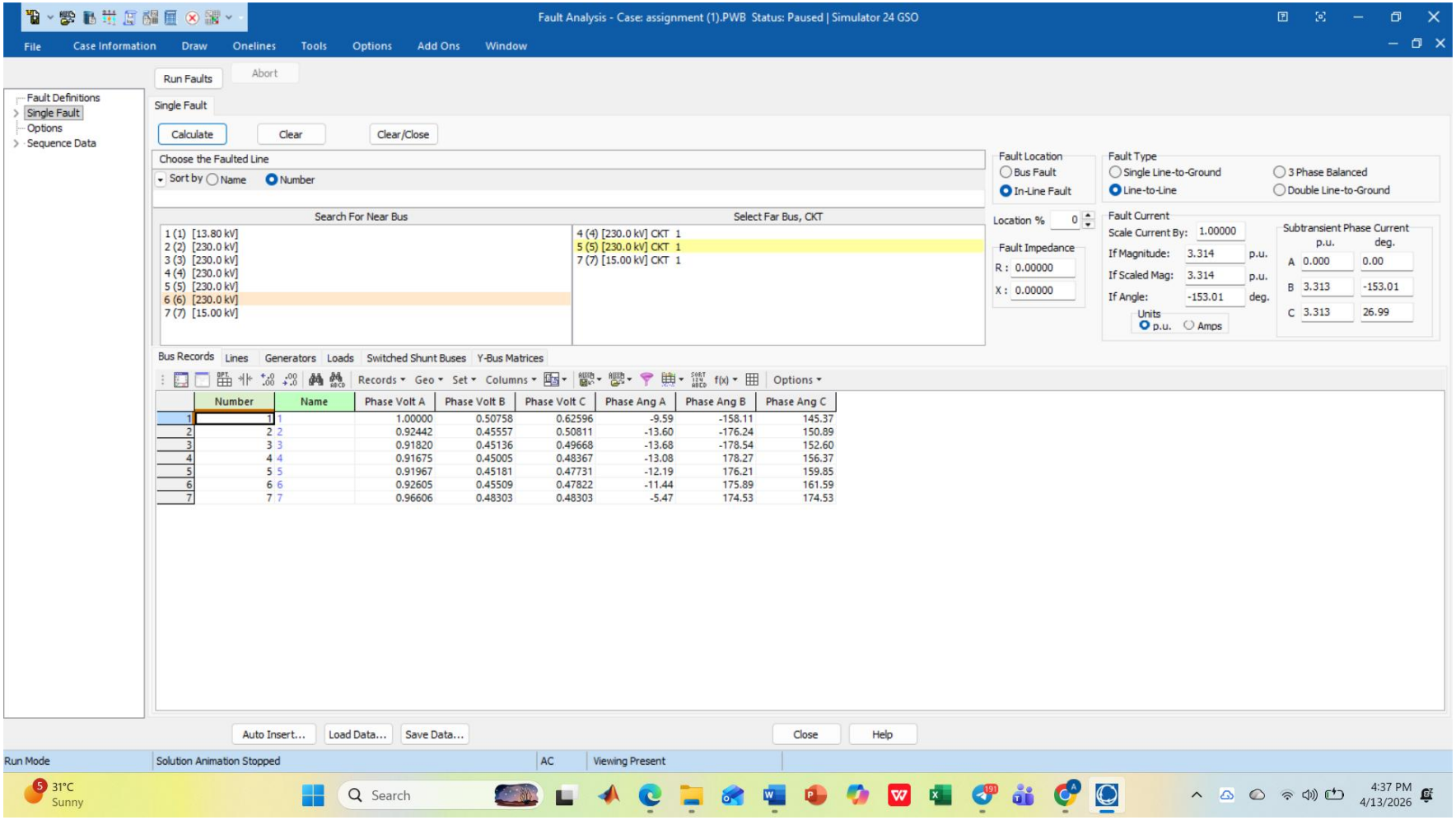
Fault Information:

Creating a Single Line to Ground Fault on Bus 2



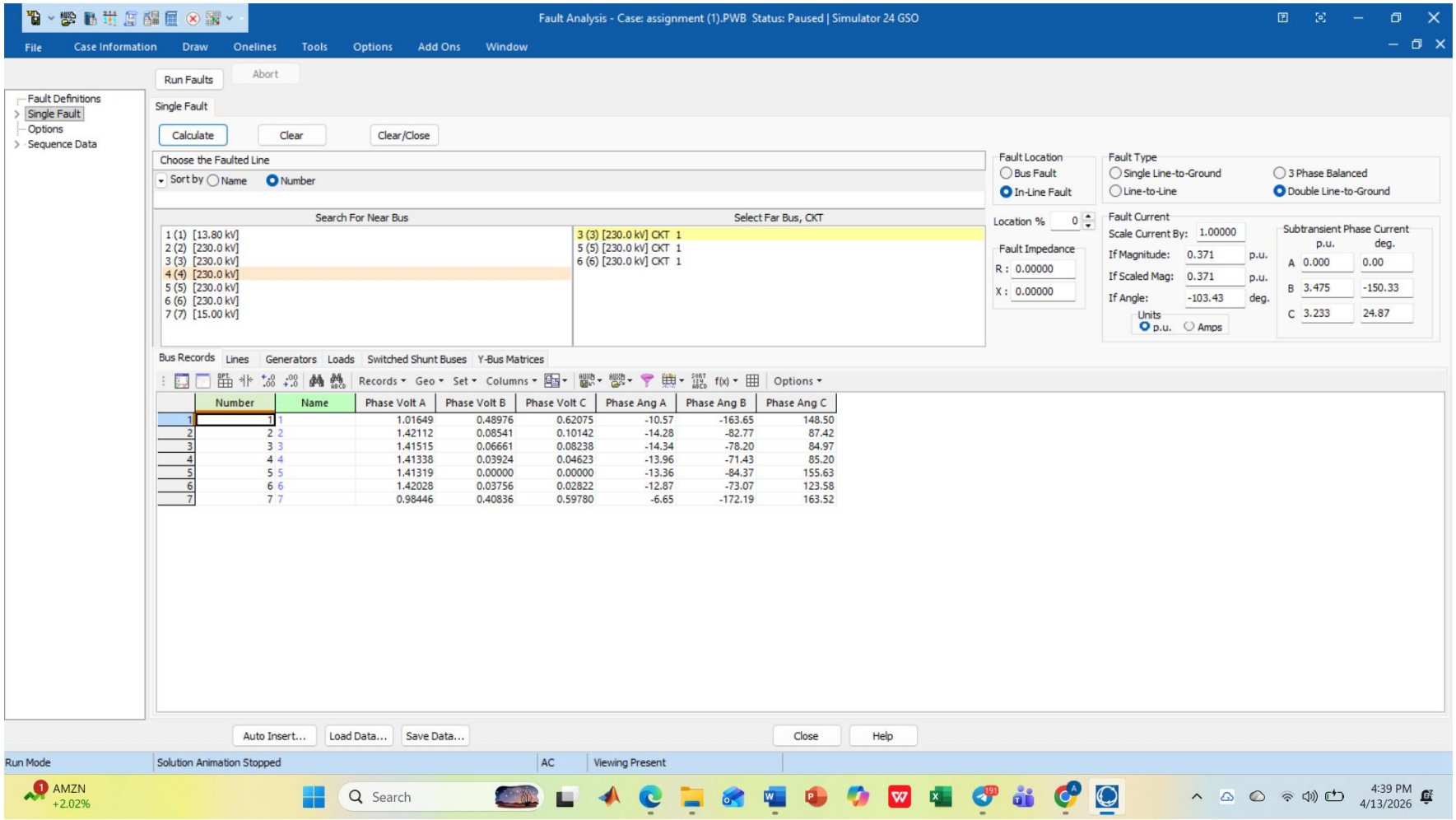
	Number	Name	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1	1	1.08249	0.98356	1.05759	-14.21	-132.85	111.09
2	2	2	0.00000	1.66566	1.79026	-27.58	-170.02	134.92
3	3	3	0.02041	1.66639	1.79051	161.50	-170.33	135.03
4	4	4	0.04446	1.66710	1.80317	145.20	-170.31	135.41
5	5	5	0.05901	1.66382	1.81595	132.27	-169.93	135.79
6	6	6	0.06358	1.66156	1.82723	119.84	-169.46	136.05
7	7	7	1.03067	0.94314	1.02056	-10.53	-128.45	114.73

1. Creating an In-Line Fault (Line to Line) Between Bus 6 and 5:

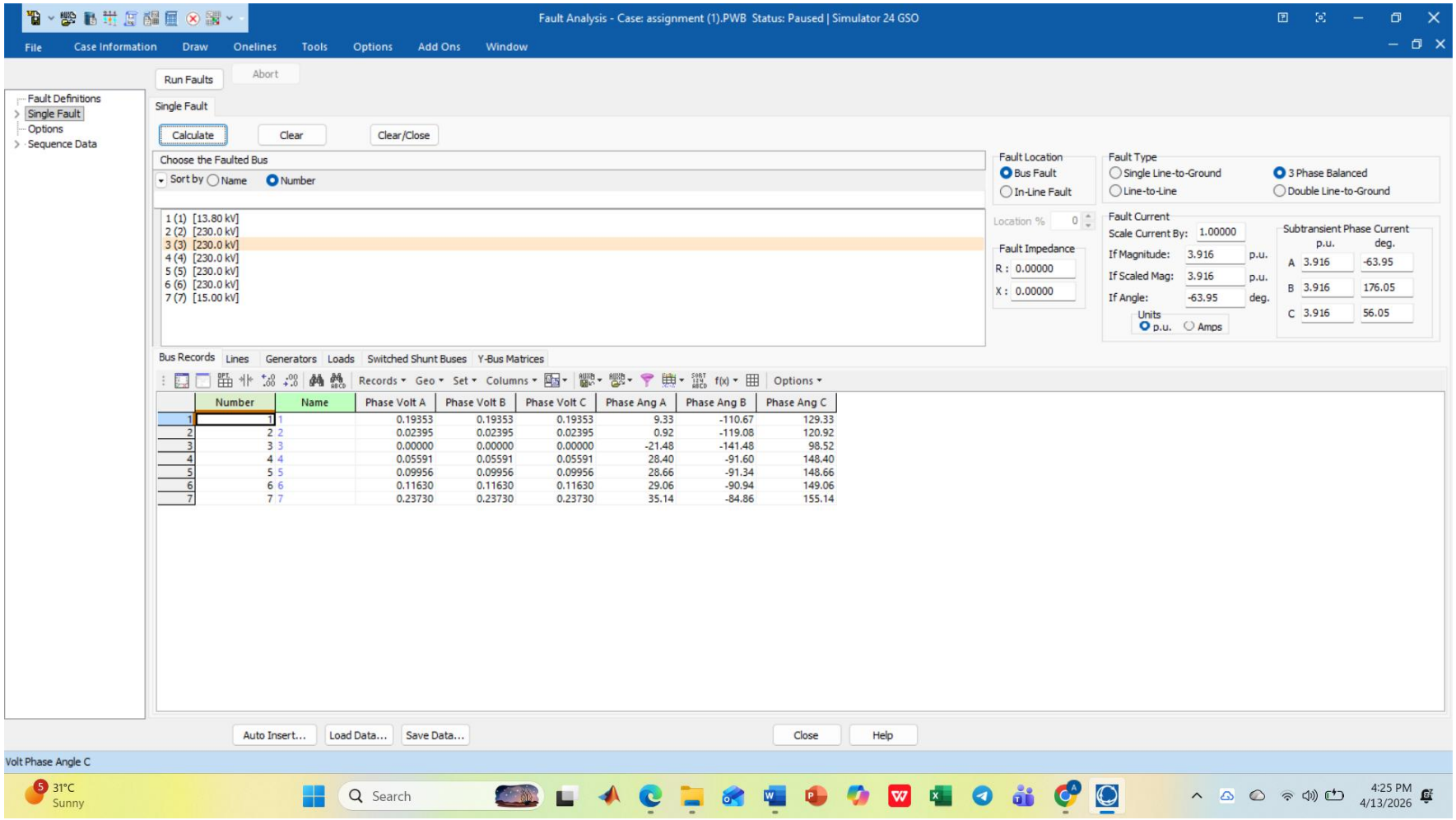


	Number	Name	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1	1	1.00000	0.50758	0.62596	-9.59	-158.11	145.37
2	2	2	0.92442	0.45557	0.50811	-13.60	-176.24	150.89
3	3	3	0.91820	0.45136	0.49668	-13.68	-178.54	152.60
4	4	4	0.91675	0.45005	0.48367	-13.08	178.27	156.37
5	5	5	0.91967	0.45181	0.47731	-12.19	176.21	159.85
6	6	6	0.92605	0.45509	0.47822	-11.44	175.89	161.59
7	7	7	0.96606	0.48303	0.48303	-5.47	174.53	174.53

2. Creating an In-Line Fault (Double Line to Ground) Between Bus 3 and 4:

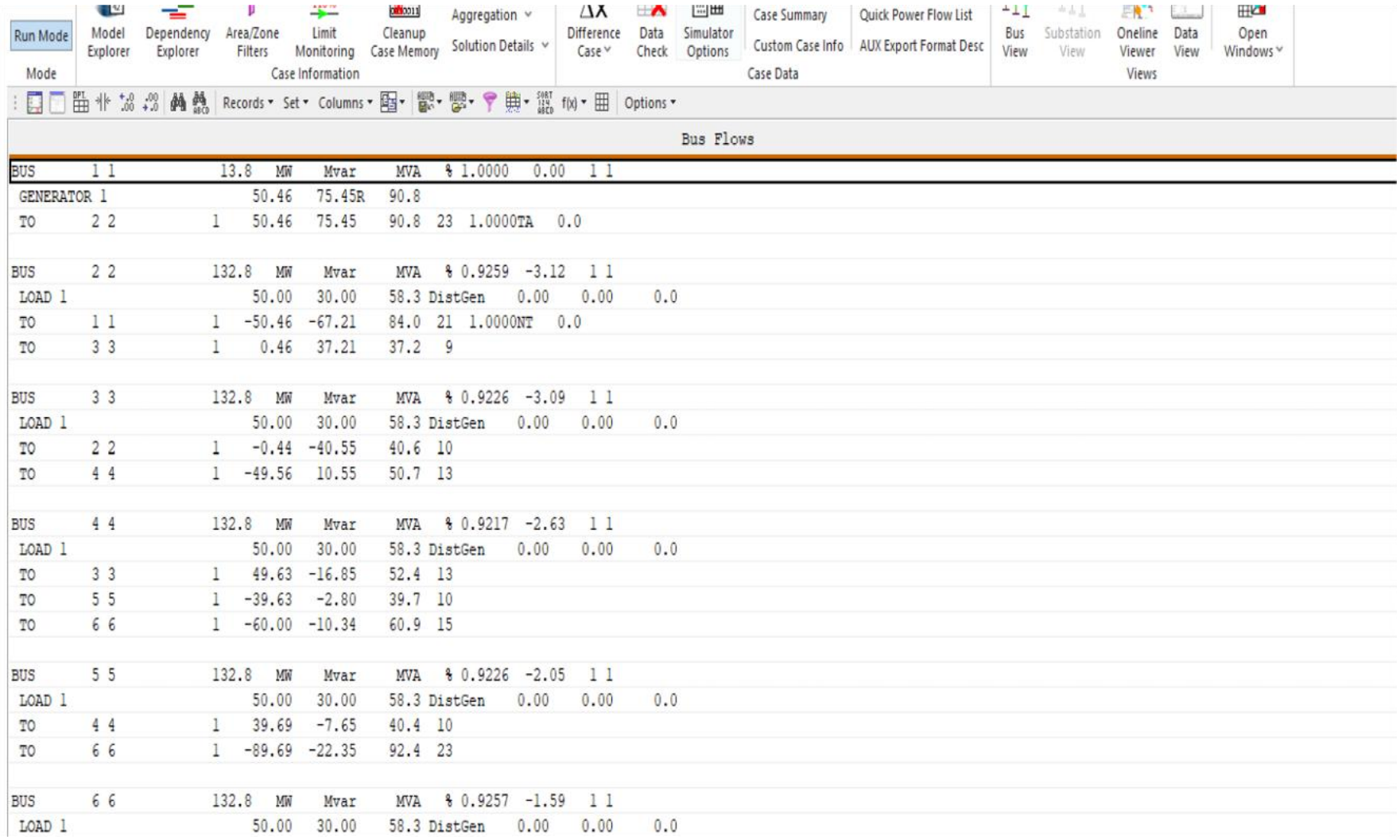


3 phase balanced fault bus



	Number	Name	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1	1	0.19353	0.19353	0.19353	9.33	-110.67	129.33
2	2	2	0.02395	0.02395	0.02395	0.92	-119.08	120.92
3	3	3	0.00000	0.00000	0.00000	-21.48	-141.48	98.52
4	4	4	0.05591	0.05591	0.05591	28.40	-91.60	148.40
5	5	5	0.09956	0.09956	0.09956	28.66	-91.34	148.66
6	6	6	0.11630	0.11630	0.11630	29.06	-90.94	149.06
7	7	7	0.23730	0.23730	0.23730	35.14	-84.86	155.14

Power flow



Bus Flows

BUS	1	1		13.8	MW	Mvar	MVA	%	1.0000	0.00	1	1
GENERATOR	1			50.46	75.45R	90.8						
TO	2	2	1	50.46	75.45	90.8	23	1.0000	TA	0.0		
BUS	2	2		132.8	MW	Mvar	MVA	%	0.9259	-3.12	1	1
LOAD	1			50.00	30.00	58.3	DistGen		0.00	0.00	0.0	
TO	1	1	1	-50.46	-67.21	84.0	21	1.0000	NT	0.0		
TO	3	3	1	0.46	37.21	37.2	9					
BUS	3	3		132.8	MW	Mvar	MVA	%	0.9226	-3.09	1	1
LOAD	1			50.00	30.00	58.3	DistGen		0.00	0.00	0.0	
TO	2	2	1	-0.44	-40.55	40.6	10					
TO	4	4	1	-49.56	10.55	50.7	13					
BUS	4	4		132.8	MW	Mvar	MVA	%	0.9217	-2.63	1	1
LOAD	1			50.00	30.00	58.3	DistGen		0.00	0.00	0.0	
TO	3	3	1	49.63	-16.85	52.4	13					
TO	5	5	1	-39.63	-2.80	39.7	10					
TO	6	6	1	-60.00	-10.34	60.9	15					
BUS	5	5		132.8	MW	Mvar	MVA	%	0.9226	-2.05	1	1
LOAD	1			50.00	30.00	58.3	DistGen		0.00	0.00	0.0	
TO	4	4	1	39.69	-7.65	40.4	10					
TO	6	6	1	-89.69	-22.35	92.4	23					
BUS	6	6		132.8	MW	Mvar	MVA	%	0.9257	-1.59	1	1
LOAD	1			50.00	30.00	58.3	DistGen		0.00	0.00	0.0	